3 Points Each

2000

1. A birthday candle stays lit for 15 minutes. For how long will 10 birthday candles stay lit if they are lit at the same time and no one blows them out?

a) 1.5 minutes
b) 15 minutes
c) 150 minutes
d) 1.5 hours
e) 15 hours

2. The little kangaroo was sick. Dr. Ohpain prescribed 3 pills for him to take one at a time every twenty minutes. How many minutes after taking the first pill will the little kangaroo take the last pill?

a) 20
b) 30
c) 40
d) 50
e) 60

3. For which of the following numbers is the product of the digits greater than the sum of the digits?

a) 112
b) 209
c) 312
d) 222
e) 211

4. Gavel lives on the second floor, and Pavel lives in the same building but has to walk up twice as many stairs as Gavel. There are no stairs to the entrance of the building.

On which floor does Pavel live?

a) on the 2nd floor
b) on the 3rd floor
c) on the 4th floor
d) on the 5th floor
e) on the 6th floor

5. Four candy bars and three lollipops cost $4.50. One candy bar costs 90 cents. How much does one lollipop cost?

a) 20 cents
b) 30 cents
c) 40 cents
d) 50 cents
e) 60 cents

6. One tour bus can seat no more than 55 people. What is the smallest number of buses needed to seat 160 people?

a) 1
b) 2
c) 3
d) 4
e) 5

7. A person needs 12 minutes to walk around a square plaza. How much time will it take for the same person to walk at the same pace around a plaza that has an area that is four times greater?

a) 48 minutes
b) 24 minutes
c) 30 minutes
d) 20 minutes
e) 36 minutes



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8. It is 120 km from Zakopane to the Krakow airport. Buses leave from Zakopane for the airport at 30 minutes past every hour. The buses drive with an average speed of 60 km per hour. A group of "Kangaroos," members of a math camp in Zakopane, is supposed to arrive at the airport at 11:30. What is the latest time their bus had to leave Zakopane to get them to the airport on time?

a) 7:30
b) 8:30
c) 9:30
d) 10:30
e) 11:30

4 Points Each

9. During the time that Kathy eats two bowls of ice cream, Betty eats three bowls of ice cream. The two girls ate 10 bowls of ice cream in one hour. How many bowls of ice cream did Kathy eat?

a) 3
b) 4
c) 5
d) 6
e) 7

10. Which four digits need to be removed from the number 4921508 to get the smallest possible three-digit number?

a) 4, 9, 2, 1
b) 4, 2, 1, 0
c) 1, 5, 0, 8
d) 4, 9, 2, 5
e) 4, 9, 5, 8

11. In each of two baskets there were 12 apples. Aria took a certain number of apples from the first basket. From the second basket, Zoe took a number of apples equal to the number of apples remaining in the first basket. How many apples were left in both baskets altogether?

a) 6
b) 12
c) 18
d) 20
e) 24

12. Students walked to the museum in rows of three. Adam, Bart, and Carl noticed that they were in the seventh row from the front and in the fifth row from the back. How many students went to the museum?

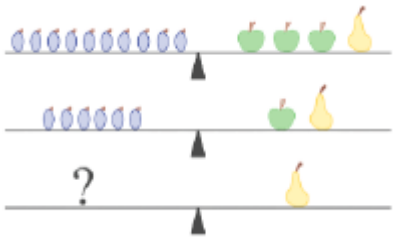
a) 12
b) 24
c) 30
d) 33
e) 36

13. Fourteen cats took part in a cat play. Some of them played moms and some of them played their kids. Every mom in this play had at least two kids. What is the greatest possible number of cat-moms in the play?

a) 3
b) 4
c) 5
d) 6
e) 7

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14. The first and the second scales are balanced (see the picture). How many plums do you need to place on the left side of the third scale to keep it in balance?



a) 2
b) 3
c) 4
d) 5
e) 6

15. Each of the five neighbors owns a rectangular plot of land with the same area. The parts of the land with flowers growing on them are fenced in (solid line in the pictures). Who has the longest fence?

a) Mr. Adam 

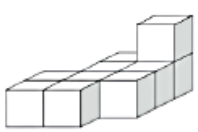
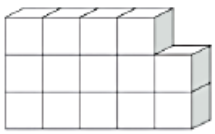
b) Mr. John 

c) Mr. Jack 

d) Mr. Peter 

e) Mr. Mark 

16. For his birthday Patrick got a box with some identical cube blocks. He used all of them to make two projects (see the picture). All the blocks together weigh 900 grams. The project on the left weighs 300 for grams and the picture shows all the blocks it is made of. How many blocks in the figure on the right are not shown?

a) 4
b) 5
c) 6
d) 7
e) 8

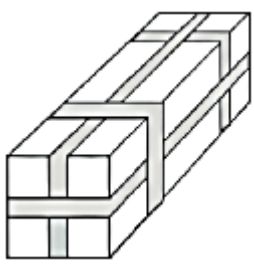
5 Points Each

17. Altogether, 6 hens eat 8 cups of grain in 3 days. How many cups of grain will 3 hens eat in 9 days?

a) 10
b) 12
c) 14
d) 16
e) 9

18. Adrianna's birthday present is placed in a box with dimensions of 10 cm x 10 cm x 30 cm and wrapped with a ribbon as shown in the picture. What is the length of the ribbon?

a) 200 cm
b) 240 cm
c) 260 cm
d) 300 cm
e) 350 cm

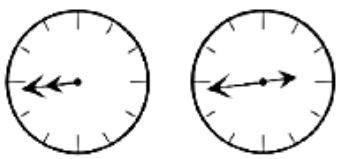


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19. Three kangaroos were born consecutively every 4 years. Right now, the oldest kangaroo is 5 times as old as the youngest one. How old is the youngest kangaroo?

a) 10
b) 8
c) 6
d) 4
e) 2

20. When Mary was leaving home between 8 and 9 o'clock in the morning, she noticed that the hour hand and the minute hand on her watch were overlapping. When she returned home between 2 and 3 o'clock in the afternoon, the hour hand and the minute hand formed a straight line (see the picture). How long was Mary away from home?

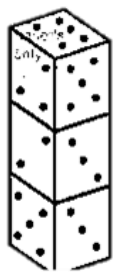


a) 5 hours
b) 5 and a half hours
c) 6 hours
d) 6 and a half hours
e) 7 hours

21. Find the number which has the following property: If we add together this number and half of this number, we will get a number which is 3 less than twice the original number.

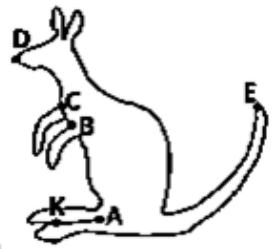
a) 2
b) 4
c) 6
d) 8
e) 10

22. Three identical dice are placed one on top of the other (see the picture). The sides which touch each other have the same numbers of dots. What is the number of dots on the bottom of the lowest die?



a) 1
b) 2
c) 3
d) 5
e) 6

Pete wanted to draw the picture of a kangaroo shown, without lifting his pencil from the paper and without going over the same line twice. At what point should he start (see the picture)?



a) A
b) B or C
c) D or E
d) K
e) There is no such point ; this is impossible.

24. A magical ball falling to the ground bounces twice as high as the height from which it was dropped. From what height was the ball dropped if it reached the height of 320 cm after the second bounce?

a) 80 cm
b) 160 cm
c) 320 cm
d) 640 cm
e) 1280 cm

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Answers key:

1. B
2. C
3. D
4. B
5. B
6. C
7. B
8. C
9. B
10. D
11. B
12. D
13. B
14. C
15. C
16. C
17. B
18. B
19. E
20. C
21. C
22. A
23. E
24. A

3 Point Solutions

1 **(B) 15 minutes**
All 10 candles will stay lit for 15 minutes since they are lit at the same time.

2 **(C) 40**
Look at the timeline below. The little kangaroo will take the last pill 20 minutes + 20 minutes = 40 minutes after taking the first pill.



3 **(D) 222**

Number	112	209	312	222	211
Product of digits	2	0	6	8	2
Sum of digits	4	11	6	6	4

The product of the digits is greater than the sum of the digits only for the number 222.

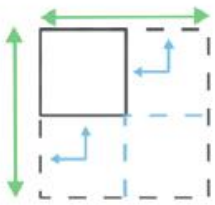
4 **(B) on the 3rd floor**
Gavel has to climb one flight of stairs to get to the second floor of the building and Pavel has to climb two flights of stairs: one to get to the second floor, and another one to get from there to the third floor. So, Pavel lives on the 3rd floor.

5 **(B) 30 cents**
$4.50 is four dollars and 50 cents. Each candy bar costs 90 cents, so four candy bars cost 3.60 dollars. Subtracting from the total price of 4.50 dollars, we have 90 cents remaining for the three lollipops, so one lollipop costs 30 cents.

6 **(C) 3**
160 = 55 + 55 + 50, so at least 3 busses are needed to seat 160 people.

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7 **(B) 24 minutes**
A big square has the area four times greater than the area of a small square. The blue arrows point to matching segments, so the perimeter of the small square is equal to the two sides (marked by green arrows) of the big square. Hence, the perimeter of the big square is twice the perimeter of the small square. Thus, the time to walk around the bigger square plaza is twice the time it takes to walk around the smaller plaza. 12 minutes are needed to walk around the smaller plaza, so 2 × 12 minutes = 24 minutes are needed to walk around the bigger plaza.



8 **(C) 9:30**
Every bus needs 2 hours to drive from Zakopane to the Krakow airport since 2 hr × 60 km per hour = 120 km, which is the distance from Zakopane to the Krakow airport. To be at the airport at 11:30, the bus had to leave Zakopane no later than 9:30.

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4 Point Solutions

9 **(B) 4**
Kathy eats two bowls of ice cream during the time in which Betty eats three bowls of ice cream. So for each 5 bowls of ice cream the girls eat, Kathy eats 2 bowls. If both girls ate 10 = 2 × 5 bowls of ice cream, then Kathy ate 4 = 2 × 2 bowls of ice cream.

10 **(D) 4, 9, 2, 5**
To have the smallest possible three-digit number, select the smallest digit from 49215, which is 1, for the hundreds digit. For the tens digit, select the smallest digit from 50 which is 0, so the ones digit is 8. The removed digits are 4, 9, 2, 5.

11 **(B) 12**
However many apples Aria takes out from her basket is exactly how many apples will remain in Zoe's basket, so 12 apples were left in both baskets altogether.

12 **(D) 33**
The boys were in the seventh row from the front and the fifth row from the back, so there were six rows before them and four rows behind them. Thus, there are 11 rows with three students in each row. Hence, 11 × 3 = 33 students went to the museum.



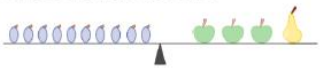
13 **(B) 4**
If 5 (or more) cats were playing moms and each mom had 2 (or more) kids, then the number of all cats would be more than 14 since 5 + 5 × 2 = 15. The greatest possible number of cat-moms in the play is 4 as shown below.



14 **(C) 4**
According to the second scale:



6 plums, one apple, and one pear can be removed from the first scale:



and this scale will stay in balance.

Here is the new first scale:



It has two identical groups of plums on the left side and two apples on the right side, so

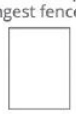


stays in balance. This allows us to remove two plums and one apple from the second scale and gives us the solution:



15 **(C) Mr. Jack**
Figures (B), (D), and (E) have the same perimeters as (A) as shown by matching arrows (blue is used for horizontal segments and green is used for vertical segments). For (C) vertical matchings are shown but the extra horizontal segments of (C) have no match with horizontal segments of the outer square, so Mr. Jack has the longest fence.

(A) Mr. Adam



(B) Mr. John



(C) Mr. Jack



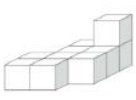
(D) Mr. Peter



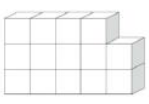
(E) Mr. Mark



16 **(C) 6**
The figure on the left weighs 300 grams and consists of 10 blocks, so each block weighs 30 grams. Both figures weigh 900 grams and use identical blocks, so the second figure weighs 600 grams and consists of 20 blocks. Only 14 blocks of the second figure are visible, so 6 blocks are not shown.



10 visible blocks

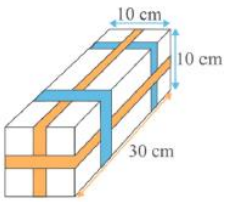


14 visible blocks

5 Point Solutions

17 **(B) 12**
6 hens eat 8 cups of grain in 3 days, so 3 hens will eat 4 cups of grain in 3 days. Hence, in 9 days, these 3 hens will eat 4 × 3 = 12 cups of grain.

18 **(B) 240 cm**
As shown in the picture, there are four pieces of ribbon in all. Two of them (shown in blue), each made of four 10 cm segments, give a total length of 80 cm (2 × 4 × 10 = 80). The other two (shown in orange) have two 10 cm segments and two 30 cm segments each, giving a total length of 160 cm [2 × (2 × 10 + 2 × 30) = 2 × 80 = 160]. This gives a total length of 80 cm + 160 cm = 240 cm.



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19 **(E) 2**

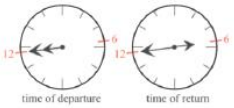
A number line will explain the relationship between ages of the two kangaroos. The age of the youngest kangaroo is represented by the orange arrow. To multiply it by 5, add 4 green arrows of the same size to it.



The black arrow represents the age of the oldest kangaroo. The difference between the ages is represented by the four green arrows, so it is 4 × the age of the youngest kangaroo. All three kangaroos were born 4 years apart, so the oldest kangaroo is 8 years older than the youngest one. This difference between their ages is always 8 years, so 8 = 4 × the age of the youngest one. Hence, right now, the youngest kangaroo is 2 years old.

20 **(C) 6 hours**

If we were to turn the clock's dial so that it shows 12:00 on the first clock (marked in red), and turn the dial on the second clock the same way, so that the minute hand is at 12 (also marked in red), then the second clock would show 6:00 as the time. Thus, the time difference is 6 hours.



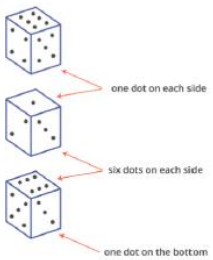
Note: The hour hand and the minute hand on the watch overlap every 1 hour, 5 minutes, and 5/11 of a minute.

21 **(C) 6**

If we add the original number and its half, then we get $\frac{3}{2}$ of the original number. If we double the original number, then we get $\frac{4}{2}$ of the original number. The difference between $\frac{4}{2}$ and $\frac{3}{2}$ is $\frac{1}{2}$ so half of the original number is 3. Therefore, the original number is 6.

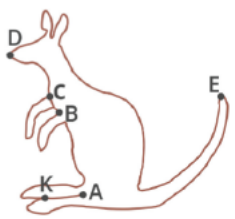
22 **(A) 1**

The faces which are touching between the middle and top dice both must show the same number, which cannot be 6, 5, or 4 visible on the top die and cannot be 2 or 3 visible on the middle die, so it must be 1. Since the dice are identical, each die has two opposite faces with 1 and 6 dots. For the middle die, the face opposite the face with 1 dot must have 6 dots, so the top face of the lowest die must also have 6 dots. Therefore, the bottom of the lowest die has 1 dot.



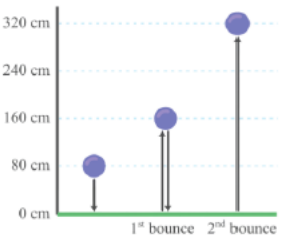
23 **(E) There is no such point; this is impossible.**

There is only one path connected to point A, so A is either the starting or the ending point of a drawing that is done without lifting the pencil from the paper or going over the same line twice. If A is the starting point, K will need the be ending point, and if A is the ending point, K will have to be the starting point, because there are two other paths that join at K that make the rest of the picture. There is another place where three paths join, which is point C. The only way we can draw three paths coming from one point is if we make it the starting or ending point, so that after coming to the point the first time and moving away from it, we can come back to the point and not have to leave it by a path that has been drawn already (starting at point C means moving away from it, coming back, and moving away again). However, if point C were the starting or ending point, then he could not draw the segment from K to A without going over that line twice. Since this means that the picture cannot be drawn this way, Pete has no starting point to draw the picture of a kangaroo shown without lifting his pencil from the paper and without going over the same line twice.



24 **(A) 80 cm**

The ball reached a height of 320 cm after the second bounce, so it must have reached a height of 320 cm ÷ 2 = 160 cm after the first bounce, and therefore would have been initially dropped from a height of 160 cm ÷ 2 = 80 cm.



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